Live Scene Understanding System

Full Project Plan — Updated Architecture (7-Tier Expanded Stack)

Wearable Narrator + Surveillance Sentinel — IIT Patna Research Project

Project	Real-time multi-modal scene understanding with dual operating modes
Modes	Wearable Narrator · Surveillance Sentinel
Architecture	7-Tier pipeline: Input → Spatial → Semantic → Audio/Multimodal → Temporal → Fused → Reasoning
Backbone	Grounded DINO 1.5 · Depth Anything V2 · RAFT/FlowFormer · DWPose/YOLOv8-P · Grounded SAM 2
New Tiers	Tier 4 — Audio & Multimodal (Whisper v3 · AudioCLIP · SigLIP/CLIP) Tier 5 — Temporal Understanding (VideoMAE · TSN/SlowFast · InternVideo2)
VLM Options	Cloud: Claude Vision / GPT-4o (~800ms) | Local: LLaVA-Next / InternVL2 (~200ms GPU) Graph: GNN over scene graph for relation-aware answers
Duration	15 weeks (Days 1–5 foundations + 13 project weeks)
Deliverable	Working system + evaluation dataset + research paper

This document is the complete technical plan for building a Live Scene Understanding System. The architecture has been expanded to a 7-tier pipeline incorporating audio/multimodal perception (Tier 4) and temporal video understanding (Tier 5), adding scene graph reasoning, optical flow, 133-keypoint pose estimation, and both cloud and local VLM options. The pipeline is built on a shared backbone so both operating modes benefit from every improvement. Mode switching uses a single runtime config flag.

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1. System Overview — 7-Tier Pipeline

The updated architecture expands the original 4-model backbone into a structured 7-tier pipeline. Each tier represents a distinct processing stage. Tiers 4 and 5 are entirely new, adding audio perception and temporal video understanding. All tiers feed into a richer unified scene dictionary.

TIER 1 — INPUT
Camera stream RGB · optional: thermal / RGBD / stereo

TIER 2 — SPATIAL PERCEPTION			
Detection Grounded DINO 1.5 open-vocab bboxes	**Depth** Depth Anything V2 monocular depth map	**Optical flow** RAFT / FlowFormer per-pixel motion vectors	**Pose** DWPose / YOLOv8-P 133 keypoints

TIER 3 — SEMANTIC UNDERSTANDING		
Segmentation Grounded SAM 2 language-conditioned masks	**Multi-object tracking** BoT-SORT / StrongSORT re-ID + occlusion-robust	**Scene graph** SGTR / RelTR subject-relation-object

TIER 4 — AUDIO & MULTIMODAL [NEW] [NEW]		
Speech recognition Whisper v3 Turbo user voice commands	**Audio events** AudioCLIP / PANNs sound class + location	**Vision-language alignment** SigLIP / CLIP semantic similarity scores

TIER 5 — TEMPORAL UNDERSTANDING [NEW] [NEW]		
Video understanding Video-LLaMA / InternVideo2 multi-frame reasoning	**Action recognition** VideoMAE v2 / X3D clip-level action labels	**Activity detection** TSN / SlowFast long-horizon activities

TIER 6 — FUSED SCENE REPRESENTATION
Unified scene dictionary objects · depth · masks · poses · flow · relations · actions · audio events · similarity scores · timestamp

TIER 7 — REASONING & NARRATION		
Cloud VLM Claude Vision / GPT-4o high quality · ~800 ms	**Local VLM** LLaVA-Next / InternVL2 offline capable · ~200 ms GPU	**Graph reasoning** GNN over scene graph relation-aware answers

OUTPUT: Wearable narrator · Surveillance sentinel · Gradio dashboard · API

Key additions vs. original plan:

- **Optical flow (RAFT/FlowFormer)** — per-pixel motion vectors added to Tier 2 for richer motion understanding.
- **Upgraded pose** — DWPose/YOLOv8-P raises keypoint count from 17 (COCO) to 133 (full body + hands + face).
- **Scene graph (SGTR/RelTR)** — generates subject-relation-object triples enabling relational queries (e.g. "person next to chair").
- **BoT-SORT / StrongSORT** — replaces ByteTrack with more occlusion-robust trackers.
- **Tier 4: Audio & Multimodal** — Whisper v3 Turbo for voice commands, AudioCLIP/PANNs for sound event detection, SigLIP/CLIP for vision-language alignment.
- **Tier 5: Temporal Understanding** — Video-LLaMA/InternVideo2 for multi-frame reasoning, VideoMAE v2/X3D for clip-level action labels, TSN/SlowFast for long-horizon activity detection.
- **Local VLM (LLaVA-Next / InternVL2)** — offline-capable inference at ~200 ms GPU latency, no API dependency.
- **Graph reasoning (GNN)** — GNN traversal over the scene graph for relation-aware answers to natural language queries.
- **Multi-modal camera input** — optional thermal, RGBD, and stereo streams alongside RGB.

2. Unified Scene Dictionary — Updated Schema

The scene dictionary is the contract between every tier and every downstream component. The updated schema adds optical flow, scene graph relations, audio events, temporal clips, and vision-language similarity scores alongside the original fields.

```
scene = {
    'timestamp': 1718123456.789,
    'frame_id':  4201,

    # ■■ Tier 2: Spatial perception ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    'objects': [
        {
            'label':    'person',
            'bbox':     [x1, y1, x2, y2],
            'conf':     0.91,
            'depth_m':  1.7,           # Depth Anything V2, median over bbox ROI
            'flow':     [dx, dy],      # NEW – RAFT/FlowFormer per-object motion
            'mask_id':  3,             # Grounded SAM 2 mask index
            'track_id': 7,             # persistent – BoT-SORT / StrongSORT
            'keypoints': [...],        # NEW – 133 keypoints (DWPose/YOLOv8-P)
        },
    ],

    # ■■ Tier 3: Semantic understanding ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    'dominant_segments':  ['road', 'sky', 'building'],
    'segment_fractions':  {'road': 0.60, 'sky': 0.22, 'building': 0.10},
    'scene_graph': [                   # NEW – SGTR / RelTR output
        {'subject': 'person', 'relation': 'sitting on', 'object': 'chair'},
        {'subject': 'cup',    'relation': 'on',          'object': 'table'},
    ],

    # ■■ Tier 4: Audio & multimodal ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    'audio_events': [                  # NEW – AudioCLIP / PANNs
        {'label': 'speech', 'conf': 0.88, 'location': 'left'},
        {'label': 'alarm',  'conf': 0.72, 'location': 'unknown'},
    ],
    'voice_command':    'navigate to exit',  # NEW – Whisper v3 Turbo
    'clip_similarity':  0.83,              # NEW – SigLIP/CLIP scene-text score

    # ■■ Tier 5: Temporal understanding ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    'action_clip':      'walking',         # NEW – VideoMAE v2 / X3D clip label
    'activity':         'commuting',       # NEW – TSN / SlowFast long-horizon
    'video_reasoning':  'Person entered from left door and sat down.',  # NEW

    # ■■ Misc ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    'free_path':        {'direction': 'forward', 'clearance_m': 2.4},
    'crowd_density':    0.34,
    'changed':          True,
    'fps':              24.1,
}
```

3. Model Choices & Rationale — Full 7-Tier Stack

Task	Model	Why not the alternative	Input → Output
Object detection	Grounded DINO 1.5	Plain YOLO limited to 80 classes; can't detect "white cane" without retraining	Text + image → bbox + class
Depth estimation	Depth Anything V2	MiDaS less accurate at edges; ZoeDepth slower	RGB → relative depth map
Optical flow [NEW]	RAFT / FlowFormer	Classic methods (Farneback) much noisier on modern GPUs	Frame pair → per-pixel flow vectors
Pose [upgraded]	DWPose / YOLOv8-P (133 kpts)	YOLOv8-Pose gives 17 kpts; DWPose adds hands + face (133)	Image → 133 COCO-WholeBody keypoints
Segmentation	Grounded SAM 2	SegFormer class-only; SAM 1 no video tracking	Text + video → instance masks
Tracking [upgraded]	BoT-SORT / StrongSORT	ByteTrack weaker on heavy occlusion; BoT-SORT integrates appearance	Detections → persistent track IDs
Scene graph [NEW]	SGTR / RelTR	No prior plan equivalent; enables relational queries	Image → subject-relation-object triples
Speech recognition [NEW]	Whisper v3 Turbo	Whisper v2 slower; Google STT requires cloud	Audio → text commands (~150ms)
Audio events [NEW]	AudioCLIP / PANNs	YAMNet narrower class set; PANNs 527 sound classes	Audio → class + location
VL alignment [NEW]	SigLIP / CLIP	CLIP older; SigLIP better zero-shot on dense scenes	Image + text → similarity score
Video understanding [NEW]	Video-LLaMA / InternVideo2	Single-frame VLMs miss temporal context	Frame sequence → natural language summary
Action recognition [NEW]	VideoMAE v2 / X3D	Pose heuristics miss subtle actions; VideoMAE learns from video	Clip → action label (running/falling/…)
Activity detection [NEW]	TSN / SlowFast	Short-clip models miss activities >5 s	Video window → long-horizon activity label
Cloud VLM	Claude Vision / GPT-4o	Local LLaVA sacrifices narration coherence	Scene dict + frame → text (~800ms)
Local VLM [NEW]	LLaVA-Next / InternVL2	Cloud VLM requires internet; local = offline capable	Scene dict + frame → text (~200ms GPU)
Graph reasoning [NEW]	GNN over scene graph	Flat VLM prompts miss relational structure	Scene graph → relation-aware answers
Tracking base	ByteTrack (fallback)	BoT-SORT preferred; ByteTrack for speed-critical paths	Detections → track IDs (lighter)

FPS vs accuracy trade-off (updated)

Mode	Target FPS	Model path	VLM call frequency	VLM preference
Wearable (outdoor)	20–30 FPS	YOLOv8n + Depth Small + RAFT-Small + Pose	Every scene change, max 1/sec	Local VLM preferred

Mode	Target FPS	Model path	VLM call frequency	VLM preference
Wearable (indoor)	15–25 FPS	Grounded DINO + Depth Base + Pose + RAFT	Every scene change, max 0.5/sec	Cloud or Local VLM
Sentinel (active)	10–15 FPS	Full 7-tier backbone (all models)	Every 5 s or on anomaly	Cloud VLM + GNN
Sentinel (idle)	5 FPS	YOLO only + depth on motion trigger	Every 30 s	Local VLM

4. Wearable Narrator — Updated Design

Audio perception (Tier 4) and temporal understanding (Tier 5) directly enhance the wearable mode. Voice commands via Whisper v3 replace button presses. Audio events (alarms, traffic) add a new SOUND hazard priority. Video understanding allows the narrator to reference recent actions ("a person just sat down to your right") rather than just the current frame.

Updated narration priority stack

Priority	Trigger condition	Example output	Latency target
1 — HAZARD	Object < 1.5 m approaching OR loud alarm detected (AudioCLIP)	"Stop — moving object directly ahead"	< 100 ms
2 — SOUND	Audio event confidence > 0.8 (NEW — Tier 4)	"Alarm to your left"	< 150 ms
3 — OBSTACLE	Free path clearance < 1 m	"Path blocked ahead, turn left"	< 200 ms
4 — NAVIGATION	Free path or junction detected	"You have 3 metres ahead, slight right"	< 500 ms
5 — DESCRIPTION	Scene change gate triggers, no hazard active (Cloud or Local VLM)	"A person at a desk to your right, distance 2 m"	< 2 s
6 — RELATIONAL	Scene graph updated with new relations (NEW — Tier 3)	"The person to your right is sitting on a chair"	< 2 s
7 — TEMPORAL	Video understanding detects activity change (NEW — Tier 5)	"Someone just entered from the door ahead"	< 3 s
8 — AMBIENT	No change for > 10 seconds	"Quiet corridor, no movement detected"	< 5 s

Voice command handling (NEW — Whisper v3 Turbo)

The wearable mode now accepts real-time voice commands from the user. Whisper v3 Turbo transcribes commands in ~150 ms. Commands are parsed for intent and injected into the narration priority queue.

```
# app/voice_handler.py
import whisper, queue

model = whisper.load_model('turbo')  # Whisper v3 Turbo
cmd_q = queue.Queue()

COMMAND_MAP = {
    'what is ahead':    'describe_front',
    'how far':          'report_depth',
    'any people':       'count_persons',
    'navigate to exit': 'find_exit',
    'read scene':       'full_description',
}

def handle_voice(audio_chunk):
    result = model.transcribe(audio_chunk)
    text = result['text'].strip().lower()
    for phrase, intent in COMMAND_MAP.items():
        if phrase in text:
            cmd_q.put({'intent': intent, 'raw': text})
```

```
        break
```

5. Surveillance Sentinel — Updated Design

The sentinel gains audio-triggered alerts (Tier 4), long-horizon activity detection (Tier 5), relational anomaly detection via the scene graph, and offline-capable summarisation via the Local VLM. Graph reasoning allows natural language queries over the event log.

Updated anomaly detection heuristics

Anomaly type	Detection heuristic	Notes
Running in walking zone	Optical flow magnitude > 3× baseline for 5+ frames (RAFT)	Flow-based; more reliable than bbox centroid delta
Audio alarm / shout	AudioCLIP confidence > 0.8 for "alarm" / "scream" class (NEW)	Triggers before visual evidence visible
Unusual dwell time	track_id remains in restricted zone > configurable seconds	Configurable in sentinel_config.yaml
Sudden crowd surge	crowd_density delta > 0.25 in < 10 frames	Same as original
Object left behind	Stationary bbox > 30 s with no associated person	Same as original
Person down / fallen	133-keypoint heuristic: torso y > hip y sustained > 3 s (DWPose)	More accurate than 17-keypoint version
Relational anomaly (NEW)	Scene graph: unexpected subject-relation-object triple detected (e.g. "person holding weapon")	SGTR/RelTR + GNN rule matching
Long-horizon activity (NEW)	TSN/SlowFast classifies activity as anomalous class (e.g. "loitering", "fighting")	Captures activities > 5 s that short-clip models miss
Restricted zone entry	bbox centroid inside configured polygon	Same as original

Updated event log structure

```
# Each entry written to events.jsonl
{
    'ts':              1718123456.789,
    'frame_id':        4201,
    'event':           'person_entered',
    'track_id':        7,
    'label':           'person',
    'depth_m':         1.4,
    'action_clip':     'walking',         # NEW – VideoMAE v2 clip label
    'activity':        'commuting',       # NEW – TSN/SlowFast long-horizon
    'relations':       [                  # NEW – scene graph triples
        {'subject': 'person', 'relation': 'carrying', 'object': 'bag'}
    ],
    'audio_events':    [{'label': 'footsteps', 'conf': 0.91}],  # NEW
    'anomaly':         False,
    'crowd_density':   0.34,
    'summary':         null               # filled by VLM every N seconds
}
```

6. Software Architecture — Updated Folder Structure

The folder structure expands to accommodate audio processing, temporal understanding, the scene graph module, and the local VLM inference option.

```
project/
  models/
    detector.py           # Grounded DINO 1.5
    depth_estimator.py    # Depth Anything V2
    flow_estimator.py     # NEW – RAFT / FlowFormer
    pose_estimator.py     # DWPose / YOLOv8-P (133 keypoints)
    segmentor.py          # Grounded SAM 2
    tracker.py            # BoT-SORT / StrongSORT
    scene_graph.py        # NEW – SGTR / RelTR
    speech_recogniser.py  # NEW – Whisper v3 Turbo
    audio_event.py        # NEW – AudioCLIP / PANNs
    vl_aligner.py         # NEW – SigLIP / CLIP
    video_understander.py # NEW – Video-LLaMA / InternVideo2
    action_recogniser.py  # NEW – VideoMAE v2 / X3D
    activity_detector.py  # NEW – TSN / SlowFast
  app/
    scene_builder.py      # fuses all tier outputs into scene dict
    scene_smoother.py     # EMA + deque temporal stability
    change_gate.py        # scene_changed() -> bool
    narrator.py           # wearable mode: VLM call -> TTS + voice cmd handler
    sentinel.py           # sentinel: logger + anomaly + summary
    graph_reasoner.py     # NEW – GNN over scene graph
    free_path.py          # depth + segmentation -> clearance
    vlm_router.py         # NEW – selects Cloud / Local / Graph VLM
    pipeline.py           # main loop, mode switch, threading
  configs/
    models.yaml           # model paths, device, precision
    prompts.yaml          # all VLM prompts (wearable + sentinel + graph)
    wearable.yaml         # wearable thresholds + voice command map
    sentinel.yaml         # zones, intervals, anomaly thresholds
    audio.yaml            # NEW – audio event classes, confidence thresholds
```

VLM Router — selecting Cloud / Local / Graph

```
# app/vlm_router.py
class VLMRouter:
    def __init__(self, config):
        self.cloud  = ClaudeVisionClient(config)   # ~800 ms, high quality
        self.local  = LLaVANextClient(config)      # ~200 ms, offline-capable
        self.graph  = GNNGraphReasoner(config)     # ~20 ms, relation-aware

    def route(self, scene: dict, query: str = None) -> str:
        # Relational query -> graph reasoner first
        if query and any(kw in query for kw in ['next to','holding','between','relation']):
```

```
        return self.graph.answer(scene['scene_graph'], query)
    # Offline / low-latency -> local VLM
    if scene.get('offline_mode') or scene['fps'] > 20:
        return self.local.narrate(scene)
    # Default -> cloud VLM for highest quality
    return self.cloud.narrate(scene)
```

7. Full Weekly Timeline — Updated for 7-Tier Stack

Days 1–5: Foundations

Day	Topic	Learn	Build
Day 1	PyTorch	Tensors, GPU, Datasets, DataLoader, nn.Module, forward pass	MLP on random data
Day 2	OpenCV	Read image, resize, crop, draw rectangle, webcam capture	Live webcam feed
Day 3	CNNs	Convolution, stride, padding, feature maps; ResNet overview	Small CNN on CIFAR-10
Day 4	Hugging Face	pipeline(), AutoModel, AutoProcessor; load SegFormer + Depth Anything	Run both on one image
Day 5	Performance	FPS, inference time, GPU vs CPU, latency measurement; audio I/O test	Three clean notebooks

Weeks 1–3: Per-model understanding

Week	Focus	Key tasks	Deliverable
Week 1	Object detection	Install Ultralytics · YOLOv8n inference · draw boxes · webcam detection loop · FPS counter · object counter	demos/week1_detection.mp4 Boxes + FPS + object count
Week 2	Depth + Flow	Run Depth Anything V2 · webcam depth loop · YOLO + depth · "Person 1.7m" NEW: Run RAFT on frame pairs · visualise flow vectors	demos/week2_depth_flow.mp4 Bbox + depth + flow overlay
Week 3	Segmentation + Scene graph	Run Grounded SAM 2 · overlay masks · segment fractions NEW: Run SGTR/RelTR · print subject-relation-object triples	demos/week3_seg_graph.mp4 Scene overlay + relation triples

Weeks 4–7: System integration

Week	Focus	Key tasks	Deliverable
Week 4	Software architecture	Create models/ classes · clean interfaces · scene_builder.py · define unified scene dict · refactor notebooks	All models callable; scene dict populated
Week 5	Dual-mode pipeline	MODE config flag · narrator.py + sentinel.py stubs · pipeline.py main loop · VLM Router (Cloud/Local/Graph) · prompts.yaml	Both modes running with VLM narration
Week 6	Temporal stability	deque(maxlen=30) history · EMA (0.8×old + 0.2×current) · change_gate.py · tune threshold on footage	Stable narration; no API rate-limit hits
Week 7	Pose (133 kpts) + Actions	Add DWPose/YOLOv8-P · 133 COCO-WholeBody keypoints · action heuristics: sitting/standing/walking/running/fallen NEW: "fallen" detection with 133-kpt precision	Pose overlay · action labels in scene dict

Weeks 7b–8: Audio & Temporal tiers (NEW)

Week	Focus	Key tasks	Deliverable
Week 7b	Audio & Multimodal (Tier 4)	Integrate Whisper v3 Turbo · voice command loop · AudioCLIP/PANNs audio event detection · SigLIP/CLIP vision-language alignment · wire audio events into scene dict	demos/week7b_audio.mp4 Voice commands + audio events in scene dict

Week	Focus	Key tasks	Deliverable
Week 8	Temporal Understanding (Tier 5)	Integrate VideoMAE v2/X3D · clip-level action labels · TSN/SlowFast activity detection · Video-LLaMA/InternVideo2 multi-frame narration · connect to VLM router	demos/week8_temporal.mp4 Action + activity labels; video-grounded narration

Weeks 9–10: Optimisation & UI

Week	Focus	Key tasks	Deliverable
Week 9	Performance profiling	Profile every model (YOLO ~20ms, Depth ~50ms, RAFT ~30ms, SAM ~40ms, Pose ~15ms, Audio ~25ms) · find bottleneck · multithreading per model · TensorRT for YOLO/Depth	Profiling report; threading; FPS improvement
Week 10	Gradio dashboard	Live webcam frame · FPS · scene dict JSON · narration text · sentinel log · mode toggle · audio event feed · scene graph visualisation tab	Running Gradio app with 7-tier display

Weeks 11–15: Evaluation & Research Paper

Weeks	Focus	Key tasks	Deliverable
Weeks 11–12	Dataset + evaluation	Record IIT Patna dataset (corridor, classroom, road, cafeteria, night) · annotate · compute FPS, BLEU-4, ROUGE-L, action accuracy, hallucination rate, re-ID accuracy, anomaly precision	All metrics tables complete
Weeks 13–14	Research paper writing	Methodology (Week 13) · Experiments & Results (Week 13) · Related Work + Intro (Week 14) · Abstract + Discussion + Conclusion (Week 14)	Full paper draft
Week 15	Revision + submission	Internal review · fix figures · format for ICVGIP / ICCV workshop / arXiv	Submitted paper

8. Optimisation Strategy — Updated Latency Breakdown

Profile first, optimise second. The updated stack adds optical flow, audio processing, and temporal models. Establish baselines before applying any optimisation.

Component	Typical latency	Optimisation lever	Expected gain
YOLO detection	~20 ms	TensorRT export, fp16, batch=1	~8 ms
Depth Anything	~50 ms	Small variant + TensorRT	~20 ms
RAFT/FlowFormer [NEW]	~30 ms	RAFT-Small variant; skip alternate frames	~12 ms
Grounded SAM 2	~40 ms	Keyframes only (every 3rd frame)	~30 ms eff.
DWPose / YOLOv8-P	~20 ms	Shares YOLO backbone; fp16	~12 ms
SGTR / RelTR [NEW]	~35 ms	Run on scene-change frames only	~10 ms eff.
Whisper v3 Turbo [NEW]	~150 ms	Run in separate audio thread; streaming	~80 ms
AudioCLIP/PANNs [NEW]	~25 ms	Run at 2 FPS (audio frame rate)	~10 ms
VideoMAE v2 [NEW]	~80 ms	Run on 16-frame clips every 2 s	~20 ms eff.
TSN / SlowFast [NEW]	~60 ms	Run at 1 FPS; share backbone with YOLO	~25 ms eff.
Scene builder (CPU)	~2 ms	Vectorise numpy ops	< 1 ms
Cloud VLM API	~800 ms	Call only on change gate; async; cache	~0 ms steady
Local VLM [NEW]	~200 ms	Run on GPU; quantise to int4	~120 ms
GNN graph reasoning [NEW]	~20 ms	Lightweight 2-layer GNN; CPU-capable	~10 ms
TTS synthesis	~150 ms	Streaming TTS; speak before full text	~60 ms

9. Evaluation Metrics — Updated for 7-Tier Stack

Metric	Measures	How to compute	Target
FPS	Speed	Time per frame, avg over 500 frames	> 15 FPS wearable; > 10 FPS sentinel
Latency (ms)	End-to-end delay	Frame capture → audio out timestamp	< 300 ms wearable priority 1
BLEU-4	Narration quality	nltk.translate.bleu_score on 50 frames	Establish baseline, improve
ROUGE-L	Narration recall	rouge_score library	Establish baseline
Action accuracy	Pose classification	Annotate 200 clips; compare to system	> 80% sit/stand/walk
Activity accuracy [NEW]	Long-horizon activity	Annotate 30-s clips; TSN/SlowFast	> 70% correct label
Hallucination rate	VLM correctness	Manual review of 100 narrations	< 10% hallucinations
Re-ID accuracy	Track identity consistency	ID switch rate on annotated sequences	< 5% ID switches
Scene graph precision [NEW]	Relation accuracy	Annotate 50 frames; compare triples	> 75% relation precision
Audio event F1 [NEW]	Sound detection	Annotate 20-min audio; F1 score	> 0.75 F1
Voice cmd accuracy [NEW]	Whisper accuracy	Test 100 commands; intent match rate	> 90% intent accuracy
Anomaly precision	Sentinel false positives	Annotate anomalies; compare to flags	Precision > 0.70
Flow accuracy [NEW]	Motion estimation	Compare to GT on annotated sequences	EPE < 2.5 px

10. Quick Reference — Updated Installation Commands

```
# ■■ Core spatial perception (Tier 2) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
pip install ultralytics            # YOLOv8, YOLOv8-P (pose), ByteTrack
pip install transformers           # Depth Anything V2 via HuggingFace
pip install segment-anything-2     # Grounded SAM 2
pip install groundingdino-py       # Grounded DINO 1.5

# NEW: Optical flow (Tier 2)
pip install torch torchvision      # RAFT ships with torchvision
# FlowFormer: git clone https://github.com/drinkingcoder/FlowFormer-Official

# NEW: Pose — DWPose 133 keypoints (Tier 2)
pip install dwpose                 # or use YOLOv8-Pose as fallback

# ■■ Semantic understanding (Tier 3) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
# SGTR / RelTR for scene graph
# git clone https://github.com/Scarecrow0/SGTR
# pip install -r SGTR/requirements.txt

# ■■ Audio & Multimodal (Tier 4) [NEW] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
pip install openai-whisper          # Whisper v3 Turbo
pip install pyaudio sounddevice     # Audio I/O
# AudioCLIP: git clone https://github.com/AndreyGuzhov/AudioCLIP
# PANNs: pip install panns_inference
pip install open_clip_torch         # SigLIP / CLIP via OpenCLIP

# ■■ Temporal understanding (Tier 5) [NEW] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
# VideoMAE v2: transformers >= 4.35 supports VideoMAEForVideoClassification
# X3D / SlowFast: pip install pytorchvideo
pip install pytorchvideo
# Video-LLaMA: git clone https://github.com/DAMO-NLP-SG/Video-LLaMA
# InternVideo2: pip install internvideo2  (or clone repo)

# ■■ Reasoning & Narration (Tier 7) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
pip install anthropic               # Claude Vision API
pip install openai                  # GPT-4o Vision API
# LLaVA-Next / InternVL2 (Local VLM):
# pip install llava  (or clone InternVL repo)
pip install torch-geometric         # GNN for graph reasoning

# ■■ Output & UI ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
pip install gtts pyttsx3            # TTS options
pip install gradio                  # Gradio dashboard

# ■■ Evaluation ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
pip install nltk rouge-score        # BLEU + ROUGE
pip install supervision             # bbox drawing, tracking utilities

# ■■ Performance ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
pip install tensorrt                # GPU optimisation (requires CUDA)
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Week-by-week deliverable checklist

Milestone	Deliverable
Days 1–5	3 notebooks (pytorch_basics, opencv_basics, huggingface_basics) + audio I/O test
Week 1	demos/week1_detection.mp4 — YOLO boxes, FPS, object counter
Week 2	demos/week2_depth_flow.mp4 — bbox + depth labels + RAFT flow overlay
Week 3	demos/week3_seg_graph.mp4 — scene overlay + SGTR relation triples printed
Week 4	All models callable via clean class interface; unified scene dict populated
Week 5	Both modes (wearable + sentinel) running; VLM router (Cloud/Local/Graph) live
Week 6	EMA + change gate; no flickering; no API rate-limit hits on 10-min run
Week 7	133-keypoint pose + action labels in scene dict; "fallen" anomaly detected
Week 7b	demos/week7b_audio.mp4 — voice commands + audio events in real time
Week 8	demos/week8_temporal.mp4 — action labels + long-horizon activity narration
Week 9	Profiling report; threading implemented; FPS target met
Week 10	Gradio app with mode toggle + scene graph tab + audio event feed
Weeks 11–12	IIT Patna dataset recorded + all metrics tables filled
Weeks 13–14	Full paper draft (Methodology, Results, Related Work, Abstract, Discussion)
Week 15	Submitted to arXiv / ICVGIP / ICCV workshop

The one rule that matters most: One working demo per week. Not a notebook. Not a screenshot. A recorded webcam video saved to demos/. That discipline — shipping something real every week — will teach you more research engineering than reading 50 papers without building anything.